

Chapter – 5: Exploring Forces

- Windy day – Sonali, Ragini – excited – cycling
- Summer vacation – recent beginning – **explore** – landscapes – village
- Pump air – tyres – set off
- Rode through village – Ragini said – wind pushing hard
- Sonali replied – riding against wind – push pedals harder – move faster
- Ride – long path – hilltop – some parts – rough – hard pedals – other parts – smoother
- Reach top – enjoy view – heard thunder – flashes of lightning – very beautiful – BUT – head back immediately
- Return journey – pass sheep herd – press bicycle bells – turn cycles – use handles
- Coming down – slope – realized – bicycles – moving down – great speed – without pedals
- Sonali yelled – thrilling – something pulling downhill – what it is?

What Is a Force?

- Activity –
 - Take – large cardboard box
 - Try – move box – diff. ways
- All ways – move box – apply push, pull
- This push, pull – obj. – force – science

What Can a Force Do to the Objects On Which It Is Applied?

- Experience – push, pull – daily life – all time
- Recall some experience – **analyse** them
- Activity –
 - **Think** – situations – push, pull – list them
 - Analyse situation – write effect – force

S. No.	Action	Push/Pull	Effect
1.	Friend holding moving bicycle from behind	Pull	Stop, decrease bicycle speed
2.	Hit moving ball with bat	Push	Change direction of moving ball
3.	Press inflated balloon	Push	Change shape of balloon

- What **conclude** – these ex.? – force – stop obj.? – change sp., dir., shape – obj.?
- Everyday life – many situations – force applied – open drawer, stretch rubber, fielder stop ball, kick football, roll chapati
- Force applied on obj. –
 - Obj. move from rest
 - Change sp. of obj.
 - Change dir. of motion of obj.
 - Bring change – shape of obj.
 - Cause – some or all these effects
- **A Step Further**

- Suppose – obj. – rest – is it under no force?
- It means – forces – this obj. – balance each other – learn more – higher grades

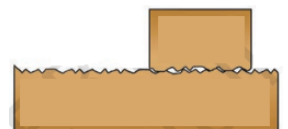
Are Forces an Interaction Between Two or More Objects?

- Push table – your hand – 1 obj. – apply force – another obj. – table
- Here – hand, table – 2 obj. – interact – each other
- Think – all actions – listed – above table – how many obj. – involve?
- **Notice** – force – result – 2 obj. – interact
- These ex. – **infer** – at least 2 obj. – interact – force in play
- **Force** – push, pull on obj. – result – obj.'s interaction – another obj.
- **SI unit** – force – **newton** – symbol – **N**
- **A Step Further**
 - Push table – hand – feel force – hand
 - Stop pushing – force – hand – disappear
 - 2 obj. interact – each obj. experience – force
 - Interaction – cease – 2 obj. – no more force

What Are the Different Types of Forces?

Contact forces

- Many situations – apply force – phy. contact – necessary b/w body, obj.
- This contact – direct – hands, body parts – OR – indirect – stick, rope
- Forces – this type – only when – phy. contact b/w obj. – **contact forces**
- **Muscular forces**
 - Example – contact force – muscular force
 - Phy. act. – walking, running, lifting, pushing, jumping, stretching – force – muscle action
 - Force result of muscle action – muscular force
 - **Muscular force** – muscles – elongate, contract – any act.
 - Animals, fish, birds, insects – muscular force – movement, survival
 - Humans – muscular force – animals – many tasks – long time
 - **Interesting fact** –
 - Muscular force – imp. role – many func. – inside body
 - This force – help – chew food – push through alimentary canal – digestion
 - Expansion, contraction – heart muscles – circulate blood – inside body – essential process – survival
- **Friction**
 - Ball – rolling – stops on own – stop pedaling – bicycle – flat road – slows and stops
 - Stops sooner – rough road – instead – smoother road
 - Learnt earlier – force – essential – change speed
 - All situations – no force – any obj. – YET – sp. decrease – eventually stop
 - Activity –
 - Take – obj. – flat base – place – table, floor
 - Gentle push – **obs.** – stop after some dist.? – any force – stop it?
 - Repeat – push obj. – opp. dir.
 - Push obj. – stop – slide some dist.



- Reason – force b/w surface – obj., table – in contact
- This force – act on obj. – opp. dir. – stop obj.
- Force – obj. – moves – another surface – **force of friction**
- Friction – opp. dir. – always – contact force – 2 surface – contact
- Friction – reason – irregularities – 2 surfaces – EVEN – look smooth
- Contact – irregularities – 2 surface – lock – oppose effort – movement
- Activity –
 - Try – above act. – again – BUT – this time – place obj. – diff. surface – glass, cloth, wood, ceramic, sand
 - Obj. – stop – travel dist.?
 - Stopping dist. – same – all surfaces?
- Diff. surface – diff. stopping dist. – force of friction – depend – nature of surface
- Friction – greater – rough surface
- **A Step Further**
 - Force of friction – only solids? – what about – liquids, gases?
 - Air, water, other liquids – force of friction – obj. through them
 - Obj. – aeroplanes, ships, boats, trains – designed – specific shapes – reduce friction

Non-contact forces

- Forces – experienced – without contact
- **Magnetic force**
 - Grade 6 – curiosity – exploring magnets – learnt – magnets
 - Learnt – magnets attract – magnetic material
 - 2 magnets – together – like poles repel – unlike poles attract
 - Earlier chapter – this book – learnt – electromagnets – behave like magnets
 - Attraction, repulsion – obj. – push, pull – force
 - Recall – magnet – apply force – another magnet, magnetic material – without contact
 - Activity –
 - Take – 2 ring magnets, wooden stick
 - Hold stick – vertical – insert – 1 magnet
 - Insert – 2nd magnet – like poles – face each other – 2nd magnet – afloat?
 - Try – gentle push – 2nd magnet – feel force?
 - Reverse poles – both magnets – 2nd magnet – still afloat?
 - Find – magnet – exert force – another magnet – without contact
 - Force by magnet – another magnet, magnetic material – **magnetic force**
 - Magnet – exert force – dist. – without contact – non-contact force
- **Electrostatic force**
 - Activity –
 - Take – plastic scale, plastic straw, polyethene, paper
 - Rub – scale, straw – polyethene
 - Don't touch – rubbed part – hand, metal obj.
 - Bring close – scale, straw – small pieces – paper – notice anything?
 - Paper pieces – pulled towards – scale, straw – stick on it – why?
 - 2 obj. – certain materials – rubbed together – elec. charge – build up – surface
 - These charges – **static charges** – no movement – themselves
 - Obj. – acquire charge – **charged object**
 - Charged obj. – apply force – uncharged obj. – without contact
 - Activity –



- Take – 2 balloons, thread, woollen cloth
 - Inflate – 2 balloons – hang them – no touching
 - Rub both – woollen cloth – release – careful – don't touch – rubbed balloon – hands – obs.
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- Balloons – move away – repelled
 - Bring – woollen cloth – close – 1 balloon – obs.
 - Balloon, cloth – move towards each other – attracted
 - Find – 2 similar charge balloons – repel – BUT – charged balloon, cloth – attract
 - This **indicate** – charge – balloon – diff. from – charge – cloth
 - Balloons – charged – same way – similar charges
 - Infer – similar charges – repel – diff. charges – attract
 - 2 kinds – static charges – positive, negative
 - Force – charged body – uncharged body, another charged body – **electrostatic** force – non-contact force
- **Gravitational force**
 - Activity –
 - Take ball – throw – vertically upward – comes down?
 - Throw again – harder – still fall back?
 - Think – diff. situations – any obj. – thrown up – any dir. – falls back
 - All obj. – fall back – earth – meaning – earth attracts them
 - Force – earth attract obj. – **gravitational force** – **force of gravity** – simply **gravity**
 - Gravitational force – without contact – non-contact force
 - Gravitation – always attractive – magnetic, electrostatic force – attractive, repulsive
 - Notice – obj. – falls down – st. vertical path
 - Obj. – thrown up – move st. – slow down – stop at top – st. vertical downward path
 - Going up – sp. decrease – obj. stops – dir. change – coming down – sp. increase
 - Obj. – **vertical motion** – movement – vert. dir. – influence – gravitational force

Weight and Its Measurement

- Force – earth pull obj. – **weight** of obj.
- Wt. – strength – earth's pull
- Wt. – force – same unit – **SI Unit** – **newton (N)**
- Activity –
 - Take – spring, few obj. – diff. masses – pencil box, tiffin box, small stone
 - Hang – 1 end – spring – nail – other end – hang obj. – obs. spring – spring stretch?
 - Hang – other obj. – notice stretch – spring – each time – same?
- Obj. – hung – spring stretch – force on obj. – earth
- Find – diff. stretch – diff. obj. – indicate – earth's pull – diff. – diff. obj.
- **A Step Further**
 - Spring balance – simple device – measure wt.
 - Consists – spring – 1 end – hook – other end
 - Hang obj. – hook – spring stretch – amt. of stretch – wt. of obj.
 - Scale on balance – marked – show wt. – newton
 - Usually – another scale – corresponding values – mass – gram (g)
 - These values – marked – assumption – spring balance – earth's gravity – attract obj.



- Let's learn – measure wt. – **spring balance** – BUT – familiarize – spring balance
- Activity –
 - Look spring balance – fig. – carefully
 - Max. wt. – measured?
- Max. wt. – measurable – 10 N – range – 0-10 N
- Activity –
 - Look – spring balance – fig. – note –
 - Wt. diff. – 2 bigger marks
 - Wt. diff. b/w – 0-01 N OR 01-02 N – 1 N
 - No. of div. – 2 bigger marks
 - 5 divisions
 - Wt. – 1 small div. – indicated wt.?
 - 1 small div. – $\frac{1N}{5} = 0.2 N$
- Use this method – calculate – smallest value – wt. – spring balance – you have
- School lab. – spring balances – diff. range – diff. div.
- Very imp. – carefully obs. – spring balance – before using
- Activity –
 - Take – spring balance, few obj. – keep mind – obj. – not heavier – max. value – wt. – spring balance – handles
 - Suspend obj. – hook – read scale – carefully – record obs.
- A Step Further**
 - Mass of obj. – measure indirectly – measure wt. – OR – compare wt. – obj. – known mass
 - Wt. of obj. – same everywhere – earth – practical purposes – acceptable – weigh obj. – measure mass
- Learnt earlier – chapter – materials – curiosity – grade 6 – mass – matter content – measured – grams (g) OR kilograms (kg) – value – same – every place
- Wt. – gravitational force – earth pulls obj. – vary – gravity – vary – BUT – mass – same
- A Step Further**
 - Table – same mass – diff. wt. – diff. places

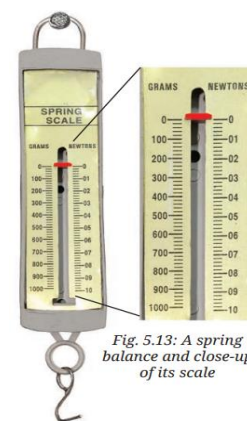


Fig. 5.13: A spring balance and close-up of its scale

	Earth	Moon	Mars	Venus	Jupiter
Mass of obj.	1 kg	1 kg	1 kg	1 kg	1 kg
Wt. of obj.	10 N	1.6 N	3.8 N	9 N	25.4 N

- A Step Further**
 - Everyday life – goods used – interested – mass – instead wt. – BUT – term wt. – used often
 - Ex. – wt. of wheat – 10 kg – BUT – scientific use – not correct – very imp. – use correct terms – correct units – even – everyday lang. – more casual

Floating and Sinking

- Take out water – bucket – use mug – mug lighter – inside water
- Activity –
 - Take – empty bottle – tightly closed lid – bucket of water
 - Push bottle in water – feel upward force? – release bottle – bounce up?
- Feel – upward push – bottle – bounce back – water surface
- Indicate – water – applies force – bottle – upward dir.
- All liquids – apply force – this force – upward dir. – **upthrust** or **buoyant force**



- Obj. – liquid – gravity – downwards – BUT – buoyant force – upwards
- Gravity > buoyant force – obj. **sink** – gravity = buoyant force – obj. **float**
- Factor – buoyant force depend – density – liquid – learn more – later chapter
- **A Step Further**
 - Archimedes – Greek scientist – discover – obj. – full, partial immersion – liquid – upward force = wt. of liquid displaced – Archimedes' Principle
 - Wt. of liquid < wt. of obj. – obj. sink – else – obj. float
- **Interesting fact –**
 - Some rocks – float – water – 1 such – Pumice – formed – volcanic eruptions
 - Lava – gas, water vapour – cools quickly – tiny bubbles – trapped inside
 - Creation – light, porous rocks – less dense – floats on water